BEYOND THE WEEK 3 BUILD

43 lessons, written out in full

Grouped by the engineering principle each one teaches, not by which sensor it happens to use. Every lesson answers the same four questions: what are we learning, what goes on the table, what do they do, and what comes out of it.

1 **Energy and trade-offs** — Gear for speed or for force — never both. — 5

2 **Control and feedback** — A robot that checks beats a robot that assumes. — 9

3 **Sensing and its limits** — Every sensor is blind to something. — 13

4 **Measurement and uncertainty** — A number is evidence, not truth. — 7

5 **Modeling and calibration** — The robot acts on what you told it, not on the world. — 5

6 **Systems integration** — Parts that each work can still fail together. — 4

G gears **U** ultrasonic **IR** infrared NEW CODE

Only 13 of the 43 need a new program. The rest run on one you have, or on **Port View**.

If you only ever run three

11 · You were the program explains why sensors exist better than any lecture can. **2 · Power is roughly conserved** is the law underneath every gear lesson they have done. **28 · Why halve it?** takes five minutes and makes the sensor stop being magic.

ALL 43, AT A GLANCE

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Each section opens with the principle itself — what it is, why it is true, and what ignoring it costs you — and then the experiments that demonstrate it. CODE marks a lesson needing a new program. The two dot columns are how hard it is to **do**, then how hard it is to **grasp**.

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Energy and trade-offs

Gear for speed or for force — never both.

THE IDEA

Your motor can only do so much. Gears let you choose how to spend it: go fast, or push hard. Ask for more pushing and you give up speed. **There is no setting that gives you both.**

WHY IT IS TRUE

Gears do not make power. They only pass along what the motor already makes — and a little gets lost as heat every time one tooth pushes another.

WHAT IT COSTS YOU IF YOU IGNORE IT

Pick gears for speed alone and you will lose every pushing match in December, and you will not know why.

THE 5 EXPERIMENTS THAT SHOW IT

1. Torque is force times radius
2. Power is roughly conserved
3. Where did the rest go?
4. Why teeth instead of a belt
5. Remote tug of war

1 Torque is force times radius

G · GEARS · HANDS ●○○ · HEAD ●●○

INTENT Turning effort equals force multiplied by radius — and the radius is a choice the builder makes, not a fact they are stuck with.

SET-UP String, a part-filled water bottle, a bare axle and a large wheel to wind onto.

THE TEST Wind the string onto a bare axle and lift the bottle straight up with the motor. Then wind the same string onto a large wheel on the same axle and lift the same bottle. Change nothing else.

OUTCOME The bare axle lifts it without complaint. The large wheel struggles or stalls outright — same motor, same bottle, same gears, five times the winding radius. **The bottle weighs the same either way, but at a bigger radius it demands far more torque.** This is why a small drive wheel pushes harder than a big one: the same physics as gearing, through a different lever.

2 Power is roughly conserved

G·U · GEARS · HANDS ●●● · HEAD ●●●

INTENT Power is force multiplied by speed, and the motor only has so much of it. Gearing slides you along that trade without creating anything — the law underneath every gear lesson they have done.

SET-UP Three gear ratios, a stopwatch, string, and a bottle you can add water to.

THE TEST For each ratio measure two things. Speed: time it over a measured meter. Pull: add water to the towed bottle until the robot can no longer move it, and record the weight. Then multiply the two numbers together for each ratio and compare the three products.

OUTCOME Speed and pull trade off almost exactly — the fast gearing is three times quicker and pulls about a third as much — and the three products land in the same region, within perhaps 30% of each other. **This is why “fast and strong” was never on the menu.** The deepest thing in the whole set; run it if you run nothing else.

3 Where did the rest go?

G · GEARS · HANDS ●○○ · HEAD ●●○

INTENT Nothing is perfectly efficient, and the energy that went missing did not vanish — it turned into heat they can feel with their hands.

SET-UP Lesson 33's numbers, and a robot you are willing to stall for 20 seconds.

THE TEST Ask why the three products in lesson 2 were close but not equal. Then stall the motor against a wall at full power for 20 seconds, switch off, and carefully touch the gears and the motor casing.

OUTCOME They are warm, sometimes properly hot — and the gear train with the most gears in it is the warmest. **Every tooth that slides against another takes a small cut.** More gears means more teeth, more cuts, and a product that comes out lower than the simple theory predicted. The discrepancy in lesson 2 now has a physical address.

4 Why teeth instead of a belt

G · GEARS · HANDS ●●○ · HEAD ●○○

INTENT Belts forgive overload and forget their position; gears remember position exactly and pass every shock to the motor. Both are deliberate engineering choices.

SET-UP The rubber belts and pulleys from the kit, plus the usual gears.

THE TEST Build the same drive twice, once with a gear pair and once with a belt and pulleys. Run each one and then grab the output wheel and hold it still.

OUTCOME The belt slips and squeals while the motor keeps turning. The gears stop dead and the motor strains. Release, and the belt drive has lost its place entirely while the gear drive has not. **Which one you want depends on whether you fear breakage or value knowing where you are.** There is no right answer here, which is itself the lesson.

5 Remote tug of war

G·IR · TWO ROBOTS · HANDS ●○○ · HEAD ●●○

INTENT Gears multiply force at the wheel; friction decides how much of it ever reaches the floor. Two different physical limits, and they can feel which one they have hit.

SET-UP Two robots, a short rope between them, a line on the floor under the middle.

THE TEST Different gearing on each. Both drivers go full power on the word. Best of three, then swap the gearings between the robots and run it again so it is not about the drivers.

OUTCOME Deep reduction wins, usually decisively. But if the winner's wheels start spinning in place without moving, gearing has stopped being the limit. **Once the tires slip, more torque buys nothing** and the answer becomes weight and rubber instead — a completely different conversation, which they will have unprompted.

PRINCIPLE 2 OF 6 · LESSONS 6–14

Control and feedback

A robot that checks beats a robot that assumes.

THE IDEA

A robot can guess what the room looks like, or it can look. **“Drive for 3 seconds” is a guess. “Drive until the sensor says 10 cm” is looking.**

WHY IT IS TRUE

If the robot never checks, it has no way of finding out it was wrong. If it does check, it can see the gap and close it — which is also why it sometimes wobbles, or stops too late.

WHAT IT COSTS YOU IF YOU IGNORE IT

This is why things that work in your classroom stop working at the competition. Different floor. Wall moved. Fresher battery.

THE 9 EXPERIMENTS THAT SHOW IT

6 Blind versus seeing
7 Keep your distance
8 Reaction distance
9 Wall follower
10 How fast can you follow?
11 You were the program
12 Piloted versus programmed
13 The overshoot curve
14 Slow and precise

6 Blind versus seeing

G·U · THE ROBOT · NEW CODE · HANDS ●●○ · HEAD ●○○

INTENT A timed drive has the room memorized. A sensor reads the room as it actually is. This is the entire reason the sensor is on the robot.

SET-UP A wall or a large book stood on end, and a floor you can mark with tape.

THE TEST Write program A: drive forward for 3 seconds, stop. Tune the seconds until it stops a hand's width from the wall. Now move the wall 30 cm further back and run the same program again. Then write program B: drive forward *until* the ultrasonic reads under 10 cm, then stop. Move the wall anywhere you like and run it again.

OUTCOME Program A stops in the middle of nowhere — or, if you moved the wall closer, hits it. Program B stops 10 cm from the wall every single time, wherever you put it. **A timed or rotation-counted drive contains a hidden assumption about how the world is arranged, and it is wrong the moment anything moves.** Worth 10 minutes on day one.

7 Keep your distance

G·U · THE ROBOT · NEW CODE · HANDS ●●○ · HEAD ●●○

INTENT Feedback control in its simplest possible form, and why a correction band that is too tight makes a robot shake instead of settle.

SET-UP A clipboard or book you hold in front of the robot and move by hand.

THE TEST Program a loop: if the distance is more than 20 cm, drive forward; if less than 20 cm, back up; otherwise stop. Run it, then walk the board slowly forwards and backwards. Then narrow the band from ±3 cm to ±1 cm and run it again.

OUTCOME The robot chases the board like a nervous dog. Wide band, it settles; narrow band, it judders endlessly, because each correction carries it straight out the other side. **That juddering is the oldest problem in control engineering and they just built it in four blocks.** The fix is not a better band but a correction that gets gentler as you approach — which is what a PID controller is.

8 Reaction distance

G·U · THE ROBOT · NEW CODE · HANDS ●●○ · HEAD ●●○

INTENT Sensing takes time and stopping takes time. Everything that happens in that gap is distance you did not ask for.

SET-UP A wall, a tape measure, and the stop-at-20 cm program from lesson 6 or 7.

THE TEST Set the program to stop when the sensor reads 20 cm. Drive at the wall from 1 meter away. Measure where the robot *actually* ends up with a real tape. Repeat at 25%, 50% and 100% motor power, three runs each.

OUTCOME It never stops at 20. At low power it may stop at 18 cm; at full power, 12 cm or closer. **Three delays stack up: the sensor samples only every few hundredths of a second, the program loop takes time to come round again, and the robot has momentum.** This is braking distance, and it is why speed limits exist.

9 Wall follower

G·U · THE ROBOT · NEW CODE · HANDS ●●● · HEAD ●●○

INTENT Overcorrection made visible: every correction is applied to a robot that has already moved somewhere else.

SET-UP A long straight wall, or a row of books lined up along the floor.

THE TEST Mount the ultrasonic pointing sideways. Drive forward continuously; if the gap to the wall is too big, steer gently towards it, if too small, steer away. Send it down a 3-meter wall.

OUTCOME It weaves down the wall in a slow S instead of tracking straight, and steering harder makes the weave worse rather than better. **Same deadband as lesson 7, except now the robot is moving, so every correction arrives late.** This is why ships zigzag and why learner drivers weave.

10 How fast can you follow?

G·U | THE ROBOT | HANDS ●●○ | HEAD ●●○

INTENT Every correction arrives late. Go faster and it arrives later still, so the weave gets wider.

SET-UP Lesson 10's wall follower and a 3-meter wall.

THE TEST Run it at 30% power. Mark on the floor where the robot gets furthest from the wall and where it gets closest, and measure between them. Repeat at 60% and 100%. **Change nothing else.**

OUTCOME The weave gets wider every time you speed it up, and at full power it may lose the wall altogether. Same program, same deadband, same correction — **the only thing that changed is how far the robot travels between one look and the next.**

11 You were the program

G·IR | THE REMOTE | NEW CODE | HANDS ●●○ | HEAD ●●●

INTENT The gap between what a driver does and what they can write down is exactly the thing a sensor puts back. If you run one lesson to explain why sensors exist, run this one.

SET-UP A simple course, a clipboard, and two students.

THE TEST One drives by remote while the other writes down every single action in order: forward two seconds, left a bit, forward, stop. Then type that exact list into the EV3 as a program and run it with nobody touching the remote.

OUTCOME It gets partway and goes wrong — drifting wide on a turn, stopping in the wrong place. The list looked complete and was not. **What is missing is everything the driver did with their eyes.** They were correcting continuously against what they saw, and never wrote it down because they never noticed doing it.

12 Piloted versus programmed

G·U·IR · BOTH SENSORS · NEW CODE · HANDS ●●○ · HEAD ●●○

INTENT Autonomy does not buy speed. It buys repeatability — which is a different virtue, and the reason competitions score an autonomous period separately.

SET-UP A short course: forward, round a box, stop in a taped square.

THE TEST Run it twice over. First a student drives by remote, timed. Then a written program does the same course, timed. Five runs of each, recording every time rather than just the best.

OUTCOME The human usually beats the program on the single fastest run. But the human's five times are spread over several seconds and the program's five are nearly identical. **Have them write down both the average and the spread — the spread is the real finding**, and it is the number most teams never think to collect.

13 The overshoot curve

G·U · THE ROBOT · NEW CODE · HANDS ●●● · HEAD ●●●

INTENT There is no best gear ratio. There is only the best ratio for a stated precision — which means the question "which gears?" cannot be answered on its own.

SET-UP Lesson 4's setup, plus two more gear ratios on the base.

THE TEST Run lesson 8 at three different gear ratios — fast, middle, deep reduction. Five runs each. For each, record the average error (how far past 20 cm it ended) and the spread (best run to worst run). Plot error against ratio on paper.

OUTCOME Fast gearing has a large error *and* a large spread — not just wrong, but wrong by a different amount each time. Deep reduction is accurate, consistent and slow. **This is the Week 3 trade-off with a number attached.** When someone asks in December which gears to use, the honest reply is: how accurately do you need to stop?

14 Slow and precise

G·U · THE ROBOT · NEW CODE · HANDS ●●○ · HEAD ●○○

INTENT Precision is purchasable and the currency is time. They have already felt the fast end of this trade; here they feel the other end.

SET-UP A wall, a real tape measure, and the deepest reduction they can build.

THE TEST Target: stop at 20 cm from the wall, judged by the tape and not by the robot. Five consecutive runs. A run is a hit if it lands between 19 and 21 cm. Then try to beat it with a faster gearing.

OUTCOME Deep reduction usually hits 5 from 5. Fast gearing scatters across 8 cm and cannot be tuned out of it however they adjust the trigger. **Notice the trigger distance they settled on is not 20** — they compensated for lesson 8's reaction distance without being told to, which is worth saying out loud to them.

PRINCIPLE 3 OF 6 · LESSONS 15–27

Sensing and its limits

Every sensor is blind to something.

THE IDEA

A sensor answers one small question and says nothing about anything else. Picking one is really about **knowing what tricks it.**

WHY IT IS TRUE

The ultrasonic sends out sound. The infrared sends out light. Sound bounces off angled things and gets soaked up by soft things. Light gets swallowed by dark colors and washed out by sunlight. They fail at completely different things.

WHAT IT COSTS YOU IF YOU IGNORE IT

A robot standing at a slight angle can be invisible. Your program will drive straight past something it is pointed right at.

THE 13 EXPERIMENTS THAT SHOW IT

15 Angle of reflection
16 The crooked robot goes blind
17 The beam is a cone
18 Two objects, one answer
19 It cannot see what is in front of it
20 Sound has a speed
21 Why a minimum range?
22 Black box, white box
23 Why color matters to one and not the other
24 Falling off with distance
25 Ambient interference
26 Sensor interference
27 Light versus sound

15 Angle of reflection

U · NOTHING · HANDS ●○○ · HEAD ●○○

INTENT Sound bounces off a flat surface exactly like light off a mirror — angle in equals angle out — so a wall at an angle is invisible.

SET-UP A flat card taped to a protractor, and the sensor on Port View at a fixed 30 cm.

THE TEST Start with the card square-on to the sensor. Tilt it five degrees at a time, reading the screen at each step. Keep going until the reading fails, and write down the angle where it broke.

OUTCOME It reads 30 cm steadily, then somewhere past roughly 15–20 degrees jumps to a large number or stops reporting. The card is still there; the sensor has lost it. **Tilt it far enough and the echo is sent off sideways, never to return.** This is how a robot drives confidently into something it is looking straight at.

16 The crooked robot goes blind

G·U · THE ROBOT · HANDS ●○○ · HEAD ●●○

INTENT The sensor does not slowly get worse as the robot turns off parallel. It works, and then it stops working completely.

SET-UP The wall follower from lesson 9, running. Your hand.

THE TEST While it is driving, nudge the robot a few degrees off parallel and watch the number on the screen. Nudge it further, five degrees at a time, until the reading dies.

OUTCOME Up to about 15 degrees crooked the number barely moves — under a centimeter on a 20 cm gap. Past about 20 degrees the echo bounces away and it reads nothing at all, so the robot decides the wall has gone and turns hard into empty floor. **The failure is a cliff, not a slope.** That is why a wall follower looks fine right up until it does not.

17 The beam is a cone

U · NOTHING · HANDS ●●○ · HEAD ●●○

INTENT The sensor does not look along a line. It shouts into a widening cone and reports the nearest thing anywhere inside it.

SET-UP A pencil standing upright, a ruler, and the sensor on Port View.

THE TEST Stand the pencil 10 cm to the *side* of the sensor's center line, 20 cm ahead. Does the sensor see it? Walk it further away along the same offset, 10 cm at a time, until it does.

OUTCOME Close in, the pencil is invisible. Further out it suddenly appears, because the beam has spread wide enough to include it; a few points let them sketch the cone on paper. **LEGO does not publish the cone angle, so this measurement is genuinely theirs.** Expect roughly 30 degrees — meaning at 50 cm it is watching a region about 25 cm across.

18 Two objects, one answer

U · NOTHING · HANDS ●○○ · HEAD ●○○

INTENT It returns one number — the first echo back — and says nothing whatsoever about anything behind it.

SET-UP Two books, at 20 cm and 40 cm, both inside the cone from lesson 17.

THE TEST Read the screen with only the far book present. Add the near book without removing the far one. Read again. Then take the *near* one away and read a third time.

OUTCOME With both present it reports 20. The far book might as well not exist — until the near one goes, when it reappears instantly. **The sensor does not see a scene; it sees one number.** In a sumo ring, the edge of the table behind an opponent is completely invisible for as long as the opponent is standing there.

19 It cannot see what is in front of it

G·U · THE ROBOT · HANDS ●○○ · HEAD ●●○

INTENT A sideways sensor answers a sideways question. It says nothing whatsoever about what is straight ahead.

SET-UP Lesson 10's wall follower, and a second wall or a box set across the end of the first.

THE TEST Let it follow the wall straight into an inside corner. **Do not stop it.** Watch the number on the screen the whole way in.

OUTCOME It holds its gap perfectly all the way to the crash, and the number never once complains — because nothing *was* wrong with the gap it was watching. **A sensor only answers the question you pointed it at.** This is the clearest argument in the set for carrying a second sensor.

20 Sound has a speed

U · NOTHING · HANDS ●○○ · HEAD ●●●

INTENT Why the sensor has a maximum range at all — and that the limit belongs to sound rather than to the electronics.

SET-UP Paper and a calculator.

THE TEST Give them 343 meters per second, the speed of sound in room-temperature air. Ask how long a 30 cm reading takes: 0.60 m of travel divided by 343. Then ask how long the maximum 250 cm reading takes.

OUTCOME 0.0017 seconds — under two thousandths of a second. The full 250 cm takes about 15 thousandths; both feel instant, but one is nine times longer. **A robot wanting 30 readings a second has only 33 thousandths to spare for each one**, so the far end of the range is bumping against the patience of sound itself.

21 Why a minimum range?

U · NOTHING · HANDS ●○○ · HEAD ●●○

INTENT It cannot hear an echo while it is still shouting. This is a law of physics rather than a flaw, and no amount of programming will fix it.

SET-UP A card and a ruler, sensor on Port View.

THE TEST Bring the card slowly in from 20 cm towards the sensor face, reading all the way. Find the exact distance at which it gives up, and write it down.

OUTCOME It tracks beautifully down to about 3 cm and then fails — freezing, or reporting something absurd. Closer than that it is simply blind. **From 3 cm the echo returns in under a fifth of a millisecond, before the outgoing burst has even finished.** The design consequence is immediate: put the sensor where nothing will ever be that close.

22 Black box, white box

U·IR | BOTH SENSORS | HANDS ●○○ | HEAD ●○○

INTENT Infrared does not measure distance. It measures reflected brightness and infers distance — and that inference breaks when the surface changes.

SET-UP A white card and a black card of the same material, at the same fixed distance.

THE TEST Fix both sensors at exactly 25 cm from the target spot. Read both sensors against the white card. Swap in the black card without moving anything. Read both again.

OUTCOME Ultrasonic reads 25 for both. Infrared reports the black card as though it were considerably further away. **Nothing moved, and yet one sensor changed its answer.** The inference inside it assumes every surface reflects alike, which is simply false — and the sensor has no way to tell you it has been fooled.

23 Why color matters to one and not the other

U·IR | BOTH SENSORS | HANDS ●○○ | HEAD ●●○

INTENT Color is a description of how a surface handles *light*. To a sound wave the word is meaningless — the cleanest proof that the two sensors live in different physical worlds.

SET-UP The black and white cards from lesson 22, cut from the same card stock.

THE TEST Same distance, same material, only the color differs. Read both sensors against both cards. Then put the real question to them: *why would sound care what color something is?*

OUTCOME Ultrasonic is identical for both. Infrared is wildly different. **Black and white card are the same object as far as the ultrasonic is concerned**, because 40 kHz sound has no mechanism for interacting with pigment at all. Students find this genuinely surprising, which is why it is worth doing as a question rather than a statement.

24 Falling off with distance

IR | THE IR SENSOR | HANDS ●●○ | HEAD ●●○

INTENT Light spreads out on the way there and again on the way back, so returning brightness collapses with distance. This is why infrared is excellent up close and hopeless far away.

SET-UP The IR sensor on Port View, a tape measure, a white card, graph paper.

THE TEST Read the proximity value at 10, 20, 30, 40, 50, 60 and 70 cm. Plot value against distance. Then do the same with the ultrasonic on the same axes.

OUTCOME The ultrasonic plot is a straight line all the way out. The infrared plot drops steeply and then flattens into uselessness past about half a meter — the last 20 cm are barely distinguishable from each other. **The two sensors are not rivals but specialists**, and the shape of these two lines is the whole explanation.

25 Ambient interference

IR · THE IR SENSOR · HANDS ●○○ · HEAD ●○○

INTENT The sensor cannot tell its own infrared from the sun's. Real systems beat this by flickering their beam at a known rate and ignoring everything that is not flickering.

SET-UP A sunny window, if the day cooperates. A desk lamp works as a fallback.

THE TEST Set the sensor a fixed distance from a card in the middle of the room and record the value. Carry the whole arrangement to a sunny window, unchanged, and record it again. Then try it with a lamp shining in from the side.

OUTCOME The reading shifts, sometimes a lot. Nothing about the card or the distance changed — only the light already in the room. **That flickering trick is also exactly how the beacon's four channels avoid each other**, so this lesson quietly explains a feature they have been using since lesson 40 without asking about.

26 Sensor interference

U·IR · TWO ROBOTS · HANDS ●○○ · HEAD ●●○

INTENT The ultrasonic assumes the only ultrasound in the room is its own echo — an assumption that is free, invisible, and wrong the moment a second robot arrives.

SET-UP Two robots with ultrasonic sensors, facing each other across a meter.

THE TEST Put both on Port View so the screens show live distance. Have both sitting still, facing each other, with nothing else nearby. Watch both numbers for 30 seconds.

OUTCOME The readings jump, drop to implausible values, or flicker. Each sensor is hearing the *other* robot's outgoing ping and timing it as its own echo. **In December there is guaranteed to be a second robot on the field.** Worth writing on the board and leaving there until then.

27 Light versus sound

U·IR · BOTH SENSORS · HANDS ●●○ · HEAD ●●○

INTENT You do not choose a sensor by what it is good at. You choose it by what fools it, and whether that thing is in your room.

SET-UP A sheet of paper ruled into two columns, and whatever the previous lessons turned up.

THE TEST Two columns: what fools the ultrasonic, what fools the infrared. Test candidates quickly — an angled card, a cushion, a black card, a shiny card, a window with sun on it, a mesh basket.

OUTCOME The two columns barely overlap. Angles and soft materials defeat the ultrasonic; color, surface finish and sunlight defeat the infrared. Almost nothing defeats both. **That near-zero overlap is the argument for carrying two sensors.** Hand this sheet back in December when they are deciding what goes on the competition robot.

PRINCIPLE 4 OF 6 · LESSONS 28–34

Measurement and uncertainty

A number is evidence, not truth.

THE IDEA

The robot never actually sees a distance. **Every number on the screen was worked out by a sum**, and it comes with some error already in it.

WHY IT IS TRUE

The sensor times an echo, then converts it using a speed it assumes. If that assumption is a little off, every reading is off — by the same amount, in the same direction, every time.

WHAT IT COSTS YOU IF YOU IGNORE IT

Take one reading and treat it as a fact, and you cannot tell a real improvement from a wobble. Teams spend December fixing things that were never broken.

THE 7 EXPERIMENTS THAT SHOW IT

28 Why halve it? 29 Two kinds of seeing 30 Build a speedometer 31 Measured pull
32 Resolution 33 Error has a shape 34 Why three trials

28 Why halve it?

U · NOTHING · HANDS ●○○ · HEAD ●●○

INTENT The sensor cannot measure distance at all. It measures *time*, and then does arithmetic. Everything else it appears to know is derived.

SET-UP A tape measure and the robot on Port View. No program.

THE TEST Ask: the sound leaves the sensor, hits the wall, comes back. *How far did the sound travel?* Let them argue it out. Then put the sensor exactly 30 cm from a wall by tape measure and read the screen.

OUTCOME The sound traveled 60 cm — 30 out and 30 back — and the screen says 30. **So the sensor must be halving it, and the tape measure proves it.** Five minutes, no equipment, and the whole sound group hangs off it. Run this one first.

29 Two kinds of seeing

U·IR · BOTH SENSORS · HANDS ●○○ · HEAD ●●○

INTENT Some sensors measure in real units. Others return an index that means nothing at all until you calibrate it against something real.

SET-UP Both sensors, one target, and the brick's Port View screen — no program needed.

THE TEST Put both sensors the same distance from the same white card. Read both values off Port View. Move the card to 20, 30, 40 cm and write down both numbers each time.

OUTCOME Ultrasonic says 20, 30, 40. Infrared says something like 18, 31, 45 — numbers that rise, but are not centimeters and do not match anything. **The infrared proximity value is a 0–100 index of reflected brightness.** Genuinely useful, but meaningless until somebody does the work of pinning it to real distances.

30 Build a speedometer

G·U · THE ROBOT · NEW CODE · HANDS ●●● · HEAD ●●○

INTENT Speed is not something the robot can detect. It is something it calculates, from two positions and a clock.

SET-UP A clear run of two meters at a wall. Nothing else.

THE TEST Loop: read the distance, wait half a second, read it again, subtract, divide by 0.5. Show the answer on the brick screen in cm per second. Drive slowly at a wall and watch the number. Then drive at full power.

OUTCOME A live speed readout that rises and falls as they drive. It only works heading straight at the wall — turn away and the number goes nonsense. **Nothing on this robot measures speed.** Every speedometer ever built does this same subtraction, including the one in your car.

31 Measured pull

G·U · GEARS · HANDS ●●○ · HEAD ●○○

INTENT Turning “it’s stronger” into a number, under a stated load.

SET-UP A loop of string, a full water bottle, and a 2-meter lane.

THE TEST Tie the bottle to the back of the robot. Run full power for exactly 5 seconds with each gear ratio, and use the sensor to record how far it got. Then add a second bottle and repeat.

OUTCOME Fast gearing moves one bottle quickly and two bottles not at all — the motor stalls and buzzes. Deep reduction drags both, slowly, without complaint. **Students will happily tell you one gearing is stronger; this makes them say by how much and under what load.** That is the difference between an opinion and a measurement.

32 Resolution

U · NOTHING · HANDS ●●○ · HEAD ●●●

INTENT Displayed precision and real accuracy are different things. Extra digits are not extra truth — a lesson worth learning long before anybody meets a spreadsheet.

SET-UP A target on a ruler so it can be slid a millimeter at a time, sensor on Port View.

THE TEST Slide the target 1 mm and read. Slide another 1 mm and read. Do ten of these, writing down the whole sequence. Then compare the total movement (10 mm) against the total change the sensor reported.

OUTCOME The number changes on nearly every slide, but not smoothly — it jumps, sometimes backwards. The ten millimeters do add up to roughly a centimeter overall. **The sensor displays to a tenth of a centimeter but is only accurate to a whole one**, so those wobbling tenths are noise wearing the costume of precision.

33 Error has a shape

U · NOTHING · HANDS ●○○ · HEAD ●●●

INTENT Random scatter and systematic bias are two different problems with two different fixes. Telling them apart is a skill that outlasts everything else in this document.

SET-UP A robot and a target, both absolutely still, plus a tape measure for the truth.

THE TEST Measure the real distance with the tape — say exactly 30 cm. Now take 20 sensor readings without touching anything, and write down all 20. Make a tally chart of the values.

OUTCOME They are not all the same, despite nothing having moved — and the tally may not sit centered on 30 either, perhaps clustering at 29.5 with every reading a little low. **Scatter is random and averaging cures it; bias is a consistent lean that averaging cannot touch.** The likeliest cause of a low bias is where you measured from: the tape starts at the bumper, the sensor sits further back. Temperature biases the other way — sound slows about 0.6 m/s per degree colder, and since the sensor still assumes 343, a cold room reads *long*.

34 Why three trials

U · NOTHING · HANDS ●○○ · HEAD ●●○

INTENT More trials buy precision and cost time. Deciding how good is good enough is an engineering judgment rather than a rule — the quiet lesson behind every table they have filled in all term.

SET-UP The 20 readings from lesson 33 and a calculator.

THE TEST Average the first 3 readings. Then average all 10. Then all 20. Write the three averages next to each other and ask how much the answer actually moved.

OUTCOME Usually very little — perhaps a couple of millimeters between the average of 3 and the average of 20. Occasionally one wild reading makes the 3-average noticeably worse, which is its own useful accident. **If three trials get within 2 mm and the task needs a centimeter, seventeen more were a waste.** Nobody tells students this, and it changes how they work.

PRINCIPLE 5 OF 6 · LESSONS 35–39

Modeling and calibration

The robot acts on what you told it, not on the world.

THE IDEA

The robot does not know how big its wheels are, or what gears you put on. **It only knows the numbers you gave it** — and those numbers are always a little bit off.

WHY IT IS TRUE

The motor counts its own turning. Turning that into centimeters on the floor needs the wheel size, the gear ratio, and no slipping. The robot cannot check a single one of them.

WHAT IT COSTS YOU IF YOU IGNORE IT

Measure the wheel 2 mm wrong and every trip is wrong, forever, and nothing on the robot will ever tell you.

THE 5 EXPERIMENTS THAT SHOW IT

35 Turning becomes traveling
36 The motor never knows
37 Stop on the line
38 Guess the ratio
39 Backlash

35 Turning becomes traveling

G·U · GEARS · HANDS ●○○ · HEAD ●○○

INTENT One multiplication is the entire bridge between turning and traveling, and every distance the robot believes it has covered rests on it.

SET-UP A ruler, a tape measure, and one standard EV3 wheel.

THE TEST Measure the wheel diameter — it should be 56 mm. Work out the circumference with π × diameter: 17.6 cm. Predict how far one full wheel rotation travels, then run exactly one rotation and measure it with the tape.

OUTCOME The prediction lands within a few millimeters; any shortfall is the wheel having slipped, which is worth naming rather than calling experimental error. **Get the wheel diameter wrong by 2 mm and every journey is wrong by 4%, forever**, with nothing anywhere on the robot to complain about it.

36 The motor never knows

G·U · GEARS · HANDS ●●○ · HEAD ●●○

INTENT The rotation encoder is watching the motor, not the world. This is the single best argument for putting the ultrasonic sensor on the robot at all.

SET-UP Three gear ratios and the rotation readout on the brick.

THE TEST Run exactly one motor rotation at each ratio. Each time record two numbers: what the motor's own rotation sensor reports, and how far the robot actually moved across the floor.

OUTCOME The motor reports 360 degrees every single time, identically. The robot travels three completely different distances. **The encoder has no idea what gears you fitted, whether the wheels slipped, or whether the robot was up on blocks going nowhere.** Only the ultrasonic looks outside the robot, and that is the point of the entire sensor unit.

37 Stop on the line

G·U | THE ROBOT | NEW CODE | HANDS ●●○ | HEAD ●●○

INTENT The sensor measures from its own face, not from the part of the robot you care about. Every reading needs an offset before it means anything useful.

SET-UP A tape line on the floor and a block or book standing on the line as a target to see.

THE TEST Challenge: stop with the robot's *front edge* exactly on the line. They will write "drive until distance < 0" and discover it cannot. Have them measure the gap between the sensor face and the front bumper — call it the offset — and stop at that number instead.

OUTCOME First attempts stop short or run the block over. Once they subtract the offset, it lands on the line; fast gearing still overshoots by a centimeter or two. **Every real robot has this problem — the camera is never where the gripper is.** Converting between where the sensor is and where the robot is is a permanent tax, not a beginner's mistake.

38 Guess the ratio

G·U | GEARS | HANDS ●●○ | HEAD ●●○

INTENT You can work out what is inside a machine purely from how it behaves on the outside.

SET-UP Two teams, spare gears, and the measurement program from Week 3.

THE TEST Team A swaps the gear pair where team B cannot see, and hands the robot over. Team B runs it for exactly one motor rotation, measures how far it traveled, and divides by the wheel circumference (17.6 cm for the standard wheel) to get wheel rotations. The ratio is one divided by that.

OUTCOME Team B gets the ratio right to within a tooth or two, without ever opening the gearbox. If they are badly out, the wheels were slipping — which is itself the finding. **Inferring hidden structure from visible behavior is not a party trick; it is most of science.** Nobody has ever seen inside the sun either.

39 Backlash

G GEARS HANDS ●○○ HEAD ●●○

INTENT Gear teeth cannot mesh perfectly or they would jam, so every mesh carries a small gap — and those gaps add up on every single reversal.

SET-UP A marker pen, a sticky label on the wheel, and no power at all.

THE TEST By hand, turn the motor shaft forwards until the wheel just begins to move, and mark the position. Now reverse the shaft and count how far it turns *before* the wheel responds. Measure it, then add a second gear pair and measure again.

OUTCOME There is a dead zone of several degrees at every reversal, and adding a gear pair roughly doubles it. The deepest reduction has the worst of it. **The deep reduction that won them accuracy in lesson 14 is quietly costing them something here** — which is a satisfying complication rather than a contradiction, and worth drawing out.

PRINCIPLE 6 OF 6 · LESSONS 40–43

Systems integration

Parts that each work can still fail together.

THE IDEA

Two things can each pass their own test and still break the moment you join them together.

WHY IT IS TRUE

Every part was built expecting certain things. Put them together and those expectations have to match up. And if a person is driving, **the person is part of the machine too.**

WHAT IT COSTS YOU IF YOU IGNORE IT

The auto program works. The driver works. The handoff between them is where you lose the match.

THE 4 EXPERIMENTS THAT SHOW IT

40 Beacon hunt
41 Search and approach
42 The handoff
43 Remote drive race

40 Beacon hunt

U·IR | BOTH SENSORS | NEW CODE | HANDS ●●● | HEAD ●●○

INTENT Different sensors answer different questions — *which way* versus *how far* — and a robot needing both answers needs both sensors.

SET-UP The IR beacon hidden somewhere in the room at robot height, switched to channel 1.

THE TEST Program the robot to spin slowly until the IR seeker's heading value is near zero, then stop and drive forward. Let it hunt. Then ask them to make it stop 15 cm short of the beacon — which they will find they cannot do with infrared alone.

OUTCOME It turns to face the beacon reliably and sets off. Stopping at a set distance is hopeless until they add the ultrasonic to do the ranging. **Neither sensor can do the other's job, and the robot needs both jobs done.** The grown-up name for combining them is sensor fusion.

41 Search and approach

G·U·IR | BOTH SENSORS | NEW CODE | HANDS ●●● | HEAD ●●○

INTENT A complete robot behavior — find, approach, stop — assembled entirely from parts they already understand. Nothing new is required.

SET-UP The beacon placed anywhere in the room, moved between runs.

THE TEST Chain three behaviors they have already built: spin until the IR heading is near zero, drive forward, stop when the ultrasonic reads under 15 cm. Place the beacon somewhere new and let it go.

OUTCOME The robot sweeps, locks on, crosses the room and parks in front of the beacon — from anywhere in the room, which is what makes it feel different from everything before it. **Three familiar pieces, combined, produce something that looks like a real robot.** A good last session before a break, and a good one to show parents.

42 The handoff

G·U·IR · BOTH SENSORS · NEW CODE · HANDS ●●● · HEAD ●●○

INTENT Humans and programs are good at different parts of the same task. Splitting the task along that line is how nearly every real robot works.

SET-UP A target object somewhere across the room, moved between runs.

THE TEST The driver pilots the robot to roughly within arm's reach of the target, then presses a button. From there a program takes over and closes to exactly 10 cm using the ultrasonic. Time the whole thing and measure the final gap.

OUTCOME Faster and more accurate than either a pure driver or a pure program. **This is the literal structure of an FTC match — an autonomous period and a driver period, each doing what it is good at.** It is also how surgical robots and the Mars rovers are operated.

43 Remote drive race

G·IR · THE REMOTE · HANDS ●○○ · HEAD ●○○

INTENT The driver is part of the machine. A robot that is hard to control is slow in practice, whatever the gear ratio promises on paper.

SET-UP A course with at least two tight turns, and a stopwatch.

THE TEST Same driver, same course, two gearings. Time the complete run including every crash, reverse and recovery — the clock does not stop when they make a mistake.

OUTCOME Fast gearing posts a quicker time in theory and a slower one in practice, because the driver overshoots turns and has to reverse. The middle gearing usually wins outright. **Any team choosing gears for December without driving them is guessing**, and this is the cheapest possible way to show that.

MATCHING A LESSON TO A STUDENT

Picking for the kid in front of you

Every lesson carries two ratings, because they come apart. **Hands** is how hard it is to actually run. **Head** is how hard the idea is. A lesson can be easy to do and hard to understand, or the other way round — and that is exactly what lets you hand two students different work without either of them noticing they got the easier one.

Needs a win today

Easy to do, easy to follow. 6 lessons.

15 Angle of reflection · 18 Two objects, one answer · 22 Black box, white box · 25 Ambient interference · 35 Turning becomes traveling · 43 Remote drive race

Good with their hands, finds the idea hard

Give them something to build or fiddle with, where the point is obvious once it works. 4 lessons.

4 Why teeth instead of a belt · 6 Blind versus seeing · 14 Slow and precise · 31 Measured pull

Gets ideas fast, finds building slow

Almost no equipment, but something real to think about. 14 lessons.

1 Torque is force times radius · 3 Where did the rest go? · 5 Remote tug of war · 16 The crooked robot goes blind · 19 It cannot see what is in front of it · 20 Sound has a speed · 21 Why a minimum range? · 23 Why color matters to one and not the other · 26 Sensor interference · 28 Why halve it? · 29 Two kinds of seeing · 33 Error has a shape · 34 Why three trials · 39 Backlash

Finished everything, wants more

Hard on both counts. 2 lessons, and they will take a full hour.

2 Power is roughly conserved · 13 The overshoot curve

The pairing that actually works

Put a **fast builder** and a **quick thinker** on the same lesson and let them split it — one runs the robot, one works out what the numbers mean. Both are needed, neither can finish alone, and **nobody has been given the baby version.**

FOR THE 14 LESSONS THAT NEED ONE

The math, in one page

29 of the 43 need no formula at all. These are the only ones that do, and every one is arithmetic this age group already has — ratio, division, mean and circumference are sixth and seventh grade material in Virginia.

Turning into distance

LESSONS 35, 36, 38

Circumference	**π × diameter**	The standard EV3 wheel is 56 mm across, so C = 17.6 cm.
Distance traveled	**wheel rotations × C**	No gearing involved yet.
With gearing	**motor rotations ÷ ratio × C**	A 3:1 reduction turns the wheel a third as far per motor rotation.
Gear ratio	**driven teeth ÷ driver teeth**	36 driven by 12 is 3:1. Idler gears in the middle do not change it.
Working backwards (38)	**ratio = motor rotations ÷ wheel rotations**	And wheel rotations = measured distance ÷ C.

Speed

LESSONS 30, 2, 12

Speed	**distance ÷ time**	The whole of lesson 30 is this one line.
From two sensor readings	**(first reading − second) ÷ seconds between**	Reading 40 cm then 25 cm half a second later is 30 cm per second.

Sound

LESSONS 28, 20, 21

Speed of sound	**343 m/s at 20 °C**	Slower by about 0.6 m/s per degree colder. The sensor still assumes 343, so a cold room reads long — the bias in lesson 33.
The round trip	**2 × distance**	Out and back. This is lesson 28 entire.
Time for a reading	**(2 × distance) ÷ 343**	30 cm gives 0.0017 s. At 3 cm it is too quick to hear — lesson 21.
What the sensor does	**(343 × time) ÷ 2**	Measure time, compute distance.

Force and power

LESSONS 1, 2

Torque	**force × radius**	Twice the winding radius needs twice the turning effort for the same load.
Power	**force × speed**	Near enough constant across gear ratios — the point of lesson 2.

Measurement

LESSONS 13, 12, 33, 34

Mean	**total ÷ how many**	All four lessons need only this.
Spread	**largest − smallest**	Lesson 12's real finding is the spread, not the average.
Bias	**mean − the tape-measure truth**	Positive or negative and consistent. Averaging will not remove it.

THE TWO EXCEPTIONS, AND ONE WARNING

Where the math runs out

Two of the 43 reach past middle school math. Neither needs to be dropped — in both cases measuring the thing is a better lesson than computing it.

17 · The beam is a cone

Computing the cone's width from its angle needs trigonometry, which is years away. **So do not compute it.** Have them mark on paper each point where the pencil first appeared, then draw a line through those marks. The drawing *is* the result, and it is a better one — they measured the cone rather than being told its angle.

24 · Falling off with distance

The real relationship has light spreading out on the way there and again on the way back, which is an inverse square twice over. **Do not derive it.** Plot the points and describe the shape in words: steep, then flat, then useless. "It drops off fast and then stops telling you anything" is the entire finding and needs no algebra at all.

One honest warning about lesson 2

You have no force meter, so "pull" gets measured as the weight the robot can still drag. That is a proxy, and the products will not come out in real units of power. **It does not matter.** The finding is that the three products are roughly equal, not what they equal — so keep the units consistent across all three ratios and compare them with each other, never against a number from a book.

BEFORE YOU BUILD A SESSION ON ANY OF THIS

Four things to check first

43 lessons is a menu, not a curriculum. Each of these could quietly spoil a group of lessons above, and all four are cheaper to find out now than in front of a class.

Bench-check 32 and 21 before you teach them

Both depend on the sensor's raw behavior right at its limits, and EV3 Classroom may round or clamp the value before it ever reaches the screen. Ten minutes at a desk with Port View tells you whether there is a lesson there or not — and if the value is clamped, 32 becomes a discussion rather than an experiment.

Lesson 2 will not come out clean, and that is the point

The three products will land within roughly 30% of each other, not on top of each other. Resist tidying that up — the gap *is* lesson 3. If a team gets a suspiciously perfect result, their wheels were almost certainly slipping during the speed run.

The infrared ones have a short working range

Proximity is useful to roughly 70 cm and the numbers crowd together well before that. Lay out 22, 23, 24 and 25 within half a meter or the effect disappears into noise. The ultrasonic lessons are far more forgiving about space.

Nothing here is timed

Most run as 15-minute stations. A few — 13, 43, 15, 24 and 2 — will take a full hour on their own, because they involve repeated trials across several configurations. Read the "the test" line before promising the class anything.

One thing this document is not

It is not a scope and sequence. The six principles have no required order, there is no assessment, and 43 lessons will not fit the term. Pick what fits the week you are actually in.